
HOMEWORK DAY 29 – – *Fundamental Theorem, Part I §4.3*

1. §4.3: 4 (verify Theorem for simple example)

2. Find $g'(x)$ where

(a) $g(x) = \int_1^x \cos(t^2) dt$

(b) $g(x) = \int_x^3 t^3 \sin t dt$

(c) $g(x) = \int_{-1}^{x^2} e^{t^2} dt$

(d) $g(x) = \int_{-1/x}^x \sqrt{1+t} dt$

(e) $g(x) = \sin(-x^2) \int_0^{x^2} e^{t^2} dt.$

3. Evaluate the following definite integrals.

(a) $\int_1^3 (9x^2 + 2x - 4) \, dx$

(b) $\int_0^4 (t^2 + t^{3/2}) \, dt$

(c) $\int_1^4 \frac{x-1}{\sqrt{x}} \, dx$

(d) $\int_0^2 (y-1)(2y+1) \, dy$

(e) $\int_1^2 \frac{s^4+1}{s^2} \, ds$

(f) $\int_1^9 \sqrt{3x} \, dx$

(g) $\int_0^{\pi/2} \sin(2x) \, dx$

(h) $\int_{\pi/3}^{\pi/2} \cos(x/2) \, dx$

(i) $\int_{-2}^2 x^2 \sin(x) \, dx$

$$(j) \int_0^{\pi/8} \sec^2(2\theta) d\theta$$

$$(k) \int_{-2}^2 f(x) dx \text{ where } f(x) = \begin{cases} 2 & \text{if } -2 \leq x \leq 0 \\ 4 - x^2 & \text{if } 0 < x \leq 2 \end{cases}$$

$$(l) \int_0^{\pi} f(x) dx \text{ where } f(x) = \begin{cases} \cos x & \text{if } 0 \leq x \leq \pi/2 \\ \sin x & \text{if } \pi/2 < x \leq \pi \end{cases}$$

HOMEWORK DAY 31 – *Indefinite integrals. Net change §4.4*

4. §4.4: 13

5. §4.4: 15

6. §4.4: 32

7. §4.4: 18

8. §4.4: 43

9. §4.4: 44

10. §4.4: 46

11. §4.4: 51

12. §4.4: 52

13. §4.4: 53

14. §4.4: 54

15. §4.4: 55

16. §4.4: 56

17. §4.4: 57

18. §4.4: 58

19. §4.4: 59

20. §4.4: 60